Revu Kavitha

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Summary

AI/ML Engineer with strong foundations in Machine Learning, Statistics, and data analysis. Experienced in building end-to-end ML pipelines, NLP systems, and AI applications using Python, Scikit-learn, and FastAPI, with a focus on solving real-world problems using structured and unstructured data.

Work Experience

Web Developer Intern

HCL Technologies

- Developed responsive UI components using React.
- Integrated frontend with backend APIs for dynamic data handling.
- Delivered features within a structured development cycle.

Student Volunteer - Debugging & Mobile App Contest

IEI Puducherry PTU

- Assisted in organizing a technical contest focused on debugging and problem-solving.
- Helped participants resolve coding issues and implementation challenges.
- Collaborated with teams to ensure smooth execution of the event.

Projects

Data Analysis & Prediction Pipeline — Live Demo | GitHub

- Built an end-to-end ML pipeline (Pandas, Scikit-learn) for churn and revenue prediction, processing 5k+ data records.
- Performed feature engineering and model tuning, improving prediction accuracy by ~10-15%.
- Generated actionable insights using Matplotlib and Seaborn for data-driven decision making.

RAG Knowledge Assistant — Live Demo | GitHub

- Developed a document-based Q&A system (FastAPI, LangChain, Vector DB) handling 100+ documents.
- Implemented semantic search using embeddings, improving retrieval relevance and reducing incorrect responses.
- Designed APIs for real-time queries with optimized response latency.

Multi-Model AI Orchestrator — Live Demo | GitHub

- Built a system (Python, API benchmarking) to compare 3+ AI models based on latency, cost, and response quality.
- Implemented evaluation metrics and scoring mechanisms for model selection.
- Enabled dynamic routing, improving response efficiency and reducing unnecessary API usage.

Deep Learning Model Lab — Live Demo | GitHub

- Developed a multimodal AI application (Hugging Face Transformers) for image-based question answering.
- Processed image inputs with optimized inference for faster response times.
- Improved model efficiency through performance tuning and pipeline optimization.

Resume Screening Platform — Live Demo | GitHub

- Built an NLP-based system (Python, Scikit-learn) to analyze and rank resumes, processing 100+ sample profiles.
- Implemented feature extraction and scoring to match resumes with job requirements.
- Generated automated insights and interview questions using structured pipelines.

Education

- **M.Tech**, Puducherry Technological University - CGPA: 9.58 (Ongoing)
- **B.Tech**, Puducherry Technological University - CGPA: 8.13
- **12th Class**, Sri Sai Junior College, Yanam - CGPA: 8.0
- **10th Class**, Dr. K. R. Narayanan Govt High School - GPA: 9.2

Achievements

- Cleared the national cutoff in Joint Entrance Examination (JEE) Mains 2020.
- Selected to represent at the Regional Level Science Exhibition (2017-18), Government of Puducherry.

Certifications

- Udemy Hadoop Certification in distributed data processing.
- Udemy SQL Bootcamp Certification in database querying and optimization.
- Udemy AI Engineer Production Track in LLM and agent deployment.
- Udemy Linux Certification in practical system administration.
- Diploma in Computer Applications, Technical Program.

Skills

- **Programming:** Python, JavaScript
- **Machine Learning:** Scikit-learn, Regression, Classification, Model Evaluation
- **AI/NLP:** NLP, LLMs, RAG
- **Data & Analysis:** Pandas, NumPy, SQL, Data Cleaning, Feature Engineering, Matplotlib, Seaborn
- **Backend:** FastAPI, REST APIs, MongoDB
- **Tools:** Jupyter Notebook, Git, Docker